

LOCI AND SETS

How to read the figures. — given --- auxiliary (added in the proof)
— what is being proved or asked

13–1 TERMINOLOGY

In this chapter we use the term *locus** (plural, *loci*), which we inherit from the days when Latin was the universal language of science. (Even in the first part of the 19th century, most mathematics was published in Latin.) As will be clear from the definition given below, a locus is just a set of points. The language may be met in later work, so we devote this short chapter to it. We also collect a number of geometric constructions in this chapter.

Definition 13–1. A locus of points is the set of all points, and only those points, which satisfy some given condition.

We may now use the language, “The locus of points P such that. . . .” This is just another way of saying, “The set of all points P such that. . . .”

**Locus* is a Latin word meaning, literally, *place* in the sense that we use the word *location*.

Examples:

(a) The *locus* of points at a given distance r from a given point is obviously a circle.

(b) The *locus* of points at a given distance d from a given line l is a pair of lines parallel to l , one on each side of l (Fig. 13-1).



FIGURE 13-1

Example (b) requires a bit of proof, which is omitted. We will discuss such proofs in the next section. This locus could also be described as the *set* of all points at distance d from l .

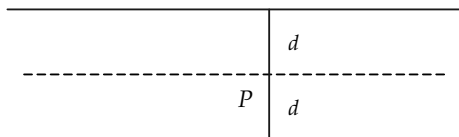


FIGURE 13-2

(c) The *locus* of points equidistant from two parallel lines is a line between the given lines.

An equivalent way of describing (c) would be to say: the *set* of all points equidistant from two parallel lines (Fig. 13-2). Again, the fact that locus (c) is a line requires proof.

Exercise Group 13-1

In the following exercises, state what you believe the locus to be. Also describe this locus as a *set*.

1. What is the locus of points P such that $\overline{PP_0} = 4$, where P_0 is a given point?
2. What is the locus of all points at a distance 1 from a line segment three units long (Fig. 13-3)? (By the distance from a point to a line segment, we mean the shortest distance from the point to any point of the segment.)
3. What is the locus of all points at a distance four units from the sides of a unit square? a square 12 units on a side? eight units?
4. A circle of radius five units is given. What is the locus of all points at a distance two units from the given circle?
5. A triangle has a fixed base four units long (Fig. 13-4). What is the locus of all possible positions of the vertex V if the area is four units? if the area is A units?

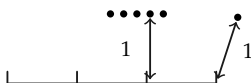


FIGURE 13-3

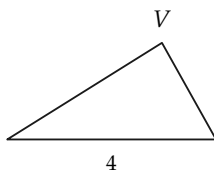


FIGURE 13-4

13-2 SOME LOCUS THEOREMS

In many locus problems the set of points can be characterized in more than one way, or we can say that two locus statements may

describe the same set of points. When this is the case, there is the following to be shown: (1) Each point of the first locus is also a point of the second locus. (The first set is a *subset* of the second set.) (2) Each point of the second locus is also a point of the first locus. (The second set is a subset of the first set.) If each is a subset of the other, the two sets are the same.

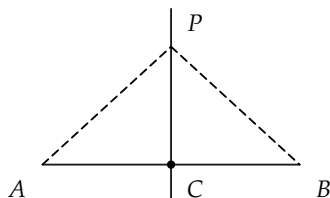


FIGURE 13-5

Theorem 13-1. *The locus of points P equidistant from two given points, A and B , is the set of points on the perpendicular bisector of the segment AB (Fig. 13-5).*

To make the proof we must show two things: (1) that if a point P is equidistant from A and B , then it is on the perpendicular bisector, and (2) that if a point is on the perpendicular bisector, then it is equidistant from A and B . These should be easy for you to prove now; in fact, both (1) and (2) were proven previously as theorems or exercises.

A brief proof will be sketched here, although your teacher may require you to write out the details. To prove (1), let C denote the midpoint of AB . Then if $P \neq C$, $\triangle PAC \cong \triangle PBC$. (Why?) Thus $\angle PCA \cong \angle PCB$, and P is on the perpendicular bisector. (How do we argue if $P = C$?) To prove (2), suppose that P is on the perpendicular bisector and $P \neq C$. Then $\triangle PCA \cong \triangle PCB$ (why?) and $PA \cong PB$. (How do we argue if $P = C$?)

A similar theorem concerns angle bisectors.

Theorem 13–2. *The locus of points interior to an angle and equidistant from the sides of the angle is the set of points on the bisector of the angle, excluding the vertex (Fig. 13–6).*

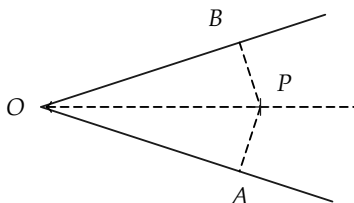


FIGURE 13–6

Proof sketch. Suppose that P is interior to the angle and equidistant from the sides. Let A and B be the feet of perpendiculars from P to the sides of the angle. (How do you know that perpendiculars from P meet the sides of the angle?) Then $\triangle POA$ and $\triangle POB$ are right triangles with two congruent sides and are congruent. Hence $\angle POA \cong \angle POB$, and OP is the angle bisector. Conversely, if P is on the bisector, then $P \neq O$, PA and PB are perpendicular to the sides of the angle, and the right triangles POA and POB are congruent. Hence $\overline{PA} = \overline{PB}$.

Theorem 13–3. *Let AB be a given segment. The set of all points P such that $\triangle APB$ is right, with AB the hypotenuse, is the circle on AB as diameter, excluding points A and B (Fig. 13–7).*

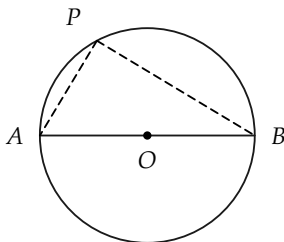


FIGURE 13–7

Let O be the midpoint of AB . It is necessary to show that if P is a point such that $AP \perp PB$, then P is on the circle whose center is O and which contains A and B .

Conversely, if P is on the circle, and different from A and B , it is necessary to show that $AP \perp PB$. But this follows from theorems on the measure of inscribed angles. (There are other proofs, too.) The details are left to the student.

Exercise Group 13-2

1. What is the set of all points that are centers of circles tangent to both sides of an angle?
2. What is the locus of all points exterior to a given circle and at a distance d from the circle?
3. A barn has dimensions 30×50 ft. A cow is tethered to a corner of the barn by a rope 60 ft long. What is the locus of points where the cow may graze?
- *4. In Fig. 13-8, points A and B are fixed. What is the locus of all points P , on a given side of the line AB , such that $\angle APB$ is 50° ?
5. What is the locus of points equidistant from three non-collinear points?
6. As in Fig. 13-9, sketch the path of a point on the rim of a wheel rolling on a line. The curve obtained is called a *cycloid*.
- *7. With the same wheel as in Exercise 6, sketch the locus if P is on a spoke of the wheel; on an extension of a spoke.
8. A small circle of radius $R/4$ rolls on the inside of a circle of radius R (Fig. 13-10). What is the path of a point on the small circle? (The curve is called a *hypocycloid*.)
- *9. What are the various possibilities that arise from Exercise 8 if the radius of the

small circle is xR ? Investigate the locus if $x = \frac{1}{2}$;

$x = \frac{1}{3}$; x is a rational number; x is an irrational number.

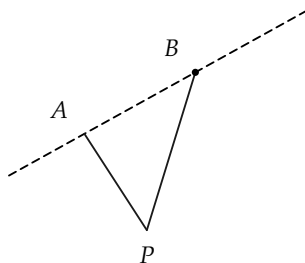


FIGURE 13-8

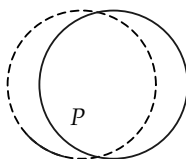


FIGURE 13-9

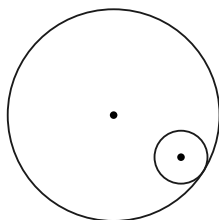


FIGURE 13-10

13-3 UNIONS AND INTERSECTIONS

Some loci are just unions or intersections of other simpler loci. To be precise, we make a definition.

Definition 13-2. Given two sets of points S_1 and S_2 , the union of the two sets is the set of all points of S_1 and S_2 . The *intersection* of S_1 and S_2 is the set of points that are common to S_1 and S_2 .

Exercise Group 13-3

1. Describe an angle as a union of two sets.
2. Describe an angle and its interior as the intersection of two sets.
3. Describe the interior of a triangle as the intersection of three sets; as the intersection of two sets.
4. Describe a triangle as the union of three sets.
5. Describe a quadrilateral as the union of four sets. Describe the interior of a simple convex quadrilateral as the intersection of two sets; as the intersection of four sets.

Theorem 13-4. *The bisectors of the angles of a triangle intersect in one point (called the incenter), which is equidistant from the three sides (Fig. 13-11).*

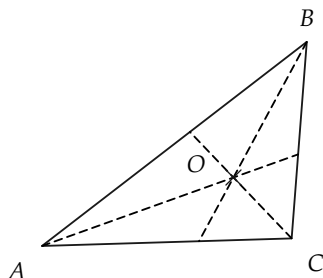


FIGURE 13-11

Proof sketch. The bisectors of angles A and B must intersect in one point O . (Why?) Then this point O is equidistant from AB and from BC . (Why?) Likewise, O is equidistant from AB and AC . Therefore, O is on the bisector of $\angle C$. (Why?) This proves the theorem.

The point O is called the *incenter* because it is the center of a circle *inscribed* in the triangle. One may also state the theorem by saying that *the bisectors of the angles are concurrent*.

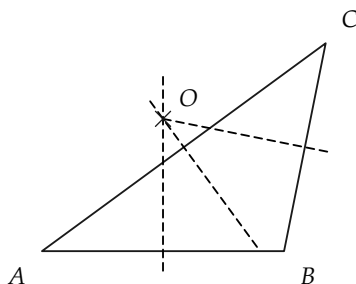


FIGURE 13-12

Theorem 13-5. *The perpendicular bisectors of the sides of a triangle intersect in one point (called the **circumcenter**), which is equidistant from the three vertices (Fig. 13-12).*

The proof is left to the student.

Exercise Group 13-4

1. Consider all lines that are parallel to the base of a triangle and that intersect the other sides in points C and D (Fig. 13-13). Show that the locus of midpoints of such segments CD is the median to the base (except for one point!).
2. Show how to inscribe a circle in a triangle. State why your construction is correct. Why is there a unique inscribed circle?
3. Show how to circumscribe a circle about a triangle. State why there is a unique circumscribing circle.
4. In Fig. 13-14, $\angle ABC$, neither right nor straight, is given. The point O is the intersection of the perpendicular bisector of AB and the line perpendicular to BC through B . Prove that there is a circle with center O containing A and B . If Q is a point of this circle and is on the same side of AB as O , prove that $\angle AQB \cong \angle ABC$.

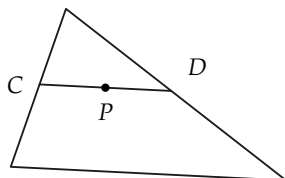


FIGURE 13-13

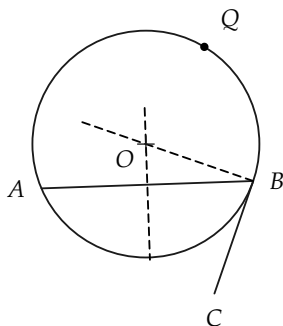


FIGURE 13-14

Theorem 13-6. *The medians of a triangle intersect in one point (called the centroid), which is two thirds of the distance from each vertex to the midpoint of the opposite side.*

Outline of proof. (Details are left as an exercise.) (1) Any two medians, say CD and AE in Fig. 13-15, intersect. Call this intersection point O . (2) Choose F and G as the midpoints of OC and OA , respectively. (3) Show that D , E , F , and G are vertices of a parallelogram. (4) Thus $AG \cong GO \cong OE$, and $FC \cong FO \cong OD$. (5) The point O has, therefore, the required distance properties. Since there is but one such point on each median, the medians are concurrent.

The centroid is the “balance point” in the following sense. A sheet of metal of uniform thickness, the shape of the triangle, would balance if supported at O . Exercise 1 of Group 13-4 may help you to see why this is so.

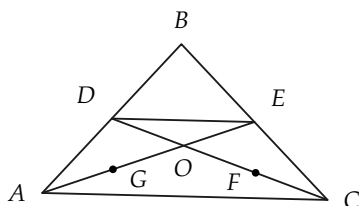


FIGURE 13-15

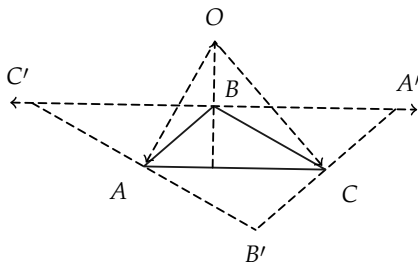


FIGURE 13-16

Theorem 13-7. *The altitudes of a triangle intersect in one point (called the orthocenter).*

Outline of proof. (Details are left as an exercise.) (1) Through each vertex, consider the line that is parallel to the opposite side of the given $\triangle ABC$. These lines will intersect to form $\triangle A'B'C'$, as in Fig. 13-16. (2) Show that $AC'BC$, $ABCB'$, and $ABA'C$ are parallelograms, and hence observe that $C'B \cong BA'$, $C'A \cong AB'$, and $B'C \cong CA'$. (3) Show that the circumcenter O of $\triangle A'B'C'$ is the point of intersection of the altitudes of $\triangle ABC$.

Exercise Group 13-5

1. Construct a triangle, given the lengths of its base, altitude, and one side.
2. Construct a triangle, given the base, the length of the altitude to this base, and one of the base angles.
- *3. As in Fig. 13-17, construct a triangle, given the lengths of two sides and an angle other than the included angle. [Hint: Use Exercise 4 in Group 13-4.]
4. Construct a triangle, given the lengths of an altitude to one side and the other two sides.
5. Construct a triangle, given

- the midpoints of the three sides.
6. Construct a triangle, given the length of the altitude to the base and the two base angles.
 7. Given a circle and a line segment AB , construct the locus of all points P such that tangents of length $\overline{AB}$ can be drawn to the circle from P .
 8. Construct a right triangle with a given segment as hypotenuse and having a side of given length.
 9. Construct a circle tangent to a given line at a given point and having a given radius.
 10. Construct the locus of all points P such that there is a circle, with given radius and center at P , which is tangent to a given circle.
 11. Construct all circles with a given radius which are tangent to two given circles of unequal radii.
 12. Construct a line tangent to a given circle and perpendicular to a given line.
 13. Given a circle and a line segment AB , construct the set of all points M such that M is the midpoint of a chord (of the circle) congruent to AB .
 14. Given an arc of a circle, construct the center of the circle.
 15. Construct an isosceles triangle, given the base, and the radius of the circumscribed circle.
 16. Construct an isosceles triangle, given the base, and the radius of the inscribed circle.
 17. Construct an isosceles triangle, given the base and the vertex angle.
 18. Construct a triangle, given two sides and the length of the median to one of these sides.
 - *19. Let P be a given point not on a given circle whose center is O . Consider all segments PX such that X is on

the circle. Show that the locus of midpoints M of these segments is a circle whose diameter is a segment on the line through P and O . Construct this circle.

*20. Given three segments, construct a triangle two of whose sides are congruent to two of these segments, and such that the median to the other side is congruent to the third segment.

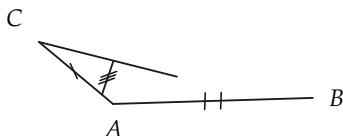


FIGURE 13-17

REVIEW OF CHAPTER 13

In earlier chapters we were using the phrases *set of points* and *geometric figure* interchangeably. In this chapter we introduced a third term, *locus*, which has exactly the same meaning as the earlier expressions.

It is important that you understand what is involved in establishing that two different “rules” determine the same locus. A statement like

the set of points having property p is the same set as that having property q

must be established by proving two things:

(1) Every point in the first set is also in the second. That is, every point having property p also has property q .

(2) Every point in the second set is also in the first. That is, every point having property q also has property p .

Review Exercises

1. If AB is a fixed segment, are the properties

p : $\angle AXB$ is acute

and

q : X is in the exterior of the circle on AB as diameter

equivalent to each other?

2. Restate Exercise 1 in conventional locus terminology.
3. If R is the set of points X such that, for a fixed segment AB , $\overline{AX} + \overline{XB} = AB$, and S is the set of points between A and B , prove that $R = S$. How else can you describe the locus of points X such that $\overline{AX} - \overline{XB} = \overline{AB}$? such that $\overline{BX} - \overline{XA} = \overline{AB}$?
4. Let AB and CD be two given segments. Describe in a second way the set of points X such that

$\angle AXB$ is obtuse and $\angle CXD$ is right;

$\angle AXB$ is obtuse or $\angle CXD$ is right;

$\angle AXB$ is acute and $\angle CXD$ is obtuse;

$\angle AXB$ is acute or $\angle CXD$ is obtuse.